MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 02 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark], [1 Mark], [1.5 Marks], [2 Marks]**). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The maximum volume of a cone that can be carved out of a solid hemisphere of radius r is: [1]

(A) $3\pi r^2$ (B) $\frac{1}{3}\pi r^3$ (C) $\frac{2}{3}\pi r^3$ (D) πr^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(B)}\ \frac{1}{3}\pi r^3}$ [0.5 Mark]

Given: Largest cone in hemisphere of radius r has base radius $= r$ and maximum height $h = r$.

Formula / Theorem Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2(r) = \boxed{\frac{1}{3}\pi \mathbf{r^3}}$

Q2. The radii of two cylinders are in the ratio 3 : 5 and their heights are in the ratio 2 : 3. The ratio of their curved surface areas is: [1]

(A) 2 : 5 (B) 5 : 2 (C) 3 : 5 (D) 6 : 25

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(A) 2 : 5}}$ [0.5 Mark]

Given: $\frac{r_1}{r_2} = \frac{3}{5}$ and $\frac{h_1}{h_2} = \frac{2}{3}$.

Formula / Theorem Used: $\frac{\text{CSA}_1}{\text{CSA}_2} = \frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \left(\frac{r_1}{r_2}\right)\left(\frac{h_1}{h_2}\right)$

$$\frac{\text{CSA}_1}{\text{CSA}_2} = \frac{3}{5} \times \frac{2}{3} = \frac{2}{5} \implies \boxed{\mathbf{2:5}}$$

Q3. If the radius of the base of a right circular cylinder is halved, keeping height the same, the ratio of the volume of reduced cylinder to the original cylinder is: [1]

(A) 1 : 2 (B) 1 : 4 (C) 1 : 8 (D) 4 : 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(B) 1 : 4}}$ [0.5 Mark]

Formula: $V_{\text{orig}} = \pi r^2 h$. For reduced cylinder ($r' = \frac{r}{2}$):

$$V_{\text{red}} = \pi\left(\frac{r}{2}\right)^2 h = \frac{1}{4}\pi r^2 h \implies \frac{V_{\text{red}}}{V_{\text{orig}}} = \boxed{\mathbf{1:4}}$$

Q4. The material of a cone is converted into the shape of a cylinder of equal radius. If the height of the cylinder is 5 cm, the height of the cone was: [1]

(A) 10 cm (B) 15 cm (C) 18 cm (D) 24 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 15 cm** [0.5 Mark]

Formula: Equating volumes with equal radius r: $\frac{1}{3}\pi r^2 h_{\text{cone}} = \pi r^2 h_{\text{cyl}}$.

$$\frac{h_{\text{cone}}}{3} = 5 \implies h_{\text{cone}} = \boxed{15 \text{ cm}}$$

Q5. **Assertion (A):** A conical vessel of base radius 5 cm and height 24 cm is full of water. If emptied into a cylindrical vessel of internal radius 10 cm, the water level rises by 2 cm.

Reason (R): The volume of water in the conical vessel is $\frac{1}{3}\pi r_1^2 h_1$, and the volume of water displaced in the cylinder is $\pi r_2^2 h_2$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: $\frac{1}{3}\pi r_1^2 h_1 = \pi r_2^2 h_2 \implies \frac{1}{3}\pi(5)^2(24) = \pi(10)^2 h_2 \implies 200\pi = 100\pi h_2 \implies h_2 = 2$ cm.
Both (A) and (R) are true, and (R) explains (A).

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. If a right circular cone and a sphere have equal radii and equal volumes, what is the ratio of the diameter of the sphere to the height of the cone? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equating Volumes [1 Mark]

Let common radius $= r$, cone height $= h$, sphere diameter $d = 2r$.

Formula / Theorem Used: $\frac{1}{3}\pi r^2 h = \frac{4}{3}\pi r^3 \implies h = 4r$

Step 2: Ratio Calculation [1 Mark]

$$\frac{\text{Diameter of Sphere}}{\text{Height of Cone}} = \frac{2r}{h} = \frac{2r}{4r} = \frac{1}{2} \implies \boxed{1:2}$$

Q7. A cone, a hemisphere, and a cylinder stand on equal circular bases of radius r and have the same height r. What is the ratio of their volumes? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Volume Formulas ($h = r$) [1 Mark]

Formula / Theorem Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \quad V_{\text{hemi}} = \frac{2}{3}\pi r^3, \quad V_{\text{cyl}} = \pi r^2 h$

Step 2: Forming Ratio [1 Mark]

With $h = r$: $V_{\text{cone}} : V_{\text{hemi}} : V_{\text{cyl}} = \frac{1}{3}\pi r^3 : \frac{2}{3}\pi r^3 : \pi r^3 = \frac{1}{3} : \frac{2}{3} : 1 = \boxed{1:2:3}$. (Multiplying by 3.)

Q8. A solid is in the shape of a right circular cone surmounted on a hemisphere, the radius of each of them being 7 cm and the height of the cone is equal to its diameter. Find the volume of the solid. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Volume Formula [1 Mark]

Radius $r = 7$ cm, diameter $= 2r = 14$ cm, cone height $h = 14$ cm.

Formula / Theorem Used: $V_{\text{solid}} = V_{\text{cone}} + V_{\text{hemi}} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2(h + 2r)$

Step 2: Substitution & Arithmetic [2 Marks]

$$V = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times (14 + 14) = \frac{154 \times 28}{3} = \frac{4312}{3} = \boxed{\mathbf{1437\frac{1}{3}}\ \text{cm}^3\ (\approx \mathbf{1437.33}\ \text{cm}^3)}$$

Q9. Two solid cones A and B are placed inside a cylindrical tube of radius 3 cm and height 21 cm. The ratio of their capacities is 2 : 1. Find the heights and capacities of the cones, and the volume of the remaining portion of the cylinder. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Heights of Cones A and B [1 Mark]

Since both cones fit collinearly inside the cylindrical tube ($r = 3$ cm, total $H = 21$ cm) sharing base radius $r = 3$ cm, their volumes are proportional to their heights: $h_1 : h_2 = 2 : 1$ with $h_1 + h_2 = 21$ cm.

$$h_1 = \frac{2}{3} \times 21 = \mathbf{14}\ \text{cm}, \qquad h_2 = \frac{1}{3} \times 21 = \mathbf{7}\ \text{cm}$$

Step 2: Capacities (Volumes) of Cones & Cylinder [2 Marks]

Formula / Theorem Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ and $V_{\text{cyl}} = \pi r^2 H$

$V_A = \frac{1}{3} \times \frac{22}{7} \times 3^2 \times 14 = 22 \times 3 \times 2 = \mathbf{132}\ \text{cm}^3$; $V_B = \frac{1}{3} \times \frac{22}{7} \times 3^2 \times 7 = \mathbf{66}\ \text{cm}^3$.

$V_{\text{cyl}} = \frac{22}{7} \times 3^2 \times 21 = 22 \times 9 \times 3 = 594\ \text{cm}^3 \implies V_{\text{rem}} = 594 - (132 + 66) = \boxed{\mathbf{396}\ \text{cm}^3}$.

Q10. A spherical glass vessel has a cylindrical neck 8 cm long and 2 cm in diameter; the diameter of the spherical portion is 8.5 cm. By measuring the amount of water it holds, a student finds its volume to be 345 cm^3. Check whether the student is correct, taking the inside measurements and $\pi = 3.14$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Dimensions & Composite Volume Formula [1.5 Marks]

Cylindrical neck: $r = \frac{2}{2} = 1$ cm, $h = 8$ cm. Spherical part: $R = \frac{8.5}{2} = 4.25$ cm.

Formula / Theorem Used: $V_{\text{vessel}} = \pi r^2 h + \frac{4}{3}\pi R^3 = \pi\left(r^2 h + \frac{4}{3}R^3\right)$

Step 2: Computing Each Part ($\pi = 3.14$) [2.5 Marks]

$V_{\text{cyl}} = 3.14 \times 1^2 \times 8 = 25.12\ \text{cm}^3$.

$V_{\text{sphere}} = \frac{4}{3} \times 3.14 \times (4.25)^3 = \frac{4}{3} \times 3.14 \times 76.765625 = \frac{964.17625}{3} \approx 321.392\ \text{cm}^3$.

$V_{\text{vessel}} = 25.12 + 321.392 = \mathbf{346.51}\ \text{cm}^3$.

Step 3: Verification Conclusion [1 Mark]

Since $346.51\ \text{cm}^3 \neq 345\ \text{cm}^3$, **Student is incorrect; correct volume $= \mathbf{346.51}\ \text{cm}^3$**.

Q11. Case Study: Architectural Marvel – The Atal Highway Tunnel

Atal Tunnel under Rohtang Pass is an engineering wonder. The cross-section of the tunnel is in the shape of a major circular segment resting on a horizontal roadway AB. The centre of the circular cross-section is O, the radius of the circular part is $R = 5\sqrt{2}$ metres, and $\angle AOB = 90°$. The total length of the tunnel is 9 km = 9000 metres. [4]

(i) Find the horizontal width of the roadway inside the tunnel (length of chord AB). [1]

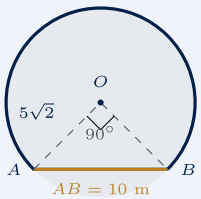
(ii) Find the total cross-sectional area of the tunnel opening. [1]

(iii) Find the perimeter of the cross-section of the tunnel. [1]

(iv) Find the surface area of the inner circular wall of the tunnel over its entire 9 km length. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Radius $R = 5\sqrt{2}$ m, minor central angle $\theta = 90°$, major sector angle $= 360° - 90° = 270°$, length $L = 9000$ m.



Formula / Theorem Used: Major Sector Area $= \dfrac{270°}{360°}\pi R^2 = \dfrac{3}{4}\pi R^2$, Major Arc $= \dfrac{3}{4}(2\pi R) = \dfrac{3}{2}\pi R$

Sub-part (i): Width of Roadway AB [1 Mark]

In right-angled $\triangle AOB$ ($\angle AOB = 90°$, $OA = OB = 5\sqrt{2}$ m):

$$AB = \sqrt{(5\sqrt{2})^2 + (5\sqrt{2})^2} = \sqrt{50 + 50} = \sqrt{100} = \boxed{\mathbf{10}\text{ m}}$$

Sub-part (ii): Cross-Sectional Area of Tunnel Opening [1 Mark]

Area = Major Sector Area + Area($\triangle AOB$) $= \frac{3}{4}\pi(5\sqrt{2})^2 + \frac{1}{2}(5\sqrt{2})(5\sqrt{2}) = \frac{75\pi}{2} + 25$.

With $\pi = \frac{22}{7}$: Area $= \frac{75}{2} \times \frac{22}{7} + 25 = \frac{825}{7} + 25 = \frac{1000}{7} \approx \boxed{\mathbf{142.86}\text{ m}^2}$ (or 142.75 m^2 using $\pi = 3.14$).

Sub-part (iii): Perimeter of Cross-Section [1 Mark]

Major Arc $= \frac{3}{2} \times \frac{22}{7} \times 5\sqrt{2} \approx \frac{15 \times 11 \times 1.414}{7} = 165 \times 0.202 = 33.33$ m.

Perimeter = Major Arc + $AB = 33.33 + 10 = \boxed{\mathbf{43.33}\text{ m}}$ (exact: $\frac{15\sqrt{2}}{2}\pi + 10$ m).

Sub-part (iv): Inner Circular Wall Surface Area Over 9000 m [1 Mark]

Wall SA = Major Arc Length $\times L = 33.335 \times 9000 \approx \boxed{\mathbf{3,00,015}\text{ m}^2}$ (or 2,99,970 m^2 using $\sqrt{2} = 1.414$).

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★